

v38 Addendum: Field Initialization and Absorbing Boundary Conditions Revisited

Linearized GR + EM PIC Sandbox, web-frontend phase

June 5, 2026

Scope. This addendum revises three sections of `gr_sandbox_v38.tex` that, while internally consistent, were found to be physically wrong or numerically broken in the course of building the interactive web demo against the v40/v41 simulator core:

1. §9.6 “Absorbing boundary layer”: the multiplicative (σdt) damping ring does not absorb the lowest standing mode of the bounded box. Replace with a derivative-friction (Klein-Gordon-like) absorber.
2. §15.9 “Field initialization”: the wave-equation convergence iteration starting from $\Phi = 0$ does not converge to a static solution for the same reason – the lowest standing mode, seeded by the initial transient, is undamped. Replace with a direct-sum 2D Liénard-Wiechert initialization that sets all six potentials analytically.
3. Add a new subsection on the DC gauge mode: under Neumann outer BC the constant-everywhere mode of a scalar potential is accelerated by any nonzero source mean and runs away; under Dirichlet the wall pins it. This was implicit in v38 but the failure mode wasn’t called out.

The discrete stability analysis of the centered-implicit friction form (which the post-step “bleed” form is *not*) is included because the result is non-trivial and easy to get wrong.

Revision 2 (2026-06-05). Corrections from running the demo end-to-end, marked inline as “Revision 2” boxes and collected in §7:

1. Field init uses a per-particle *boundary-mean subtraction* (a gauge shift) plus a *frozen-friction settle*, not the multiplicative taper of the first draft (which distorts gradients).
2. The friction absorber uses a *zero interior floor* (wall-ramp only) for dynamics: a nonzero floor damps the time-varying orbital near-field of moving sources and causes a numerical inspiral.
3. The `pic_binary` COM drift was a *TSC deposition* artifact, fixed by the *BUMP* $R = 8$ kernel – not, as first conjectured, the init transient. The near-circular orbit also needs the discrete-force velocity calibration $v_{\text{factor}} \approx 1.05$.

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1 Diagnosis of the v38 absorber

The v38 absorbing ring multiplies the leapfrog update by $(1 - \sigma_k dt)$ in a polynomial-tapered ring of depth N cells:

$$\Phi_k^{n+1} = [2\Phi_k^n - \Phi_k^{n-1} + c^2 dt^2 (\nabla^2 \Phi^n + s_c \rho_k)] (1 - \sigma_k dt). \quad (1)$$

This factor does two distinct things: (a) it dissipates outgoing wave energy, which is what we want; and (b) it pulls Φ toward zero everywhere inside the ring, which effectively imposes a Dirichlet boundary at the wall. Effect (a) only acts on field amplitude that *reaches* the ring; effect (b) is the reason the static fixed point of the box is the Dirichlet Poisson solution rather than the free-space one.

The failure mode is mode-dependent. Consider the standing modes $\phi_{mn}(x, y) = \sin(m\pi x/L) \sin(n\pi y/L)$ of a square box with zero-Dirichlet walls. The lowest mode $(m, n) = (1, 1)$ has its **antinode at the center** and **nodes at the wall**. Its overlap with the multiplicative damping array, concentrated in the outer ring, is suppressed by the eigenfunction’s small wall amplitude. Specifically, the per-step decay factor for mode ϕ_{mn} is approximately

$$1 - dt \bar{\sigma}_{mn}, \quad \bar{\sigma}_{mn} = \frac{\int_{\Omega} \sigma(x, y) \phi_{mn}^2 d^2x}{\int_{\Omega} \phi_{mn}^2 d^2x}. \quad (2)$$

For the lowest mode this average is $\mathcal{O}(N/L)$ times the wall value of σ ; for high- k modes (which oscillate with appreciable amplitude at the wall) it is $\mathcal{O}(1)$ times the wall value. At $N/L \sim 1/8$ (a typical absorber ring depth) the lowest mode is damped roughly an order of magnitude slower than high- k modes.

Empirical signature. With `pic.static` (a single stationary particle), particles frozen, and the field initialized by the v38 §15.9 convergence iteration, the value of Φ_g at the box center oscillates with period $\sim 2L/c$ and amplitude $\sim |\Phi_g^{\text{static}}|$ indefinitely. The probe trajectory shows envelope swing of $\sim 50\%$ of $|\Phi_g^{\text{static}}|$ after 4000 steps with no decay trend. The absorbing ring is not absorbing the lowest mode.

The v38 §15.9 prescription – “iterate the leapfrog with frozen particles until the field converges” – relies on the absorber to remove the transient launched by switching the source on at $t = 0$. That transient has Fourier content at every mode; the lowest component is excited as strongly as any other, and once excited it persists. The convergence iteration therefore does not, in fact, converge to the static solution; it converges to the static solution *plus* a residual standing-mode oscillation of comparable amplitude.

2 Direct-sum Liénard-Wiechert initialization

The correct initialization is to set each perturbation potential at $t = 0$ to the static Poisson solution of the deposited sources, then start the wave equation from there with zero time-derivative.

For an isolated N -particle configuration in 2D, the static solution is known analytically – it is the 2D Liénard-Wiechert potential, which under the Lorenz gauge is built from the same $\ln r$ kernel for all six fields:

$$\Phi_g(\mathbf{x}) = -\frac{s_c^g}{2\pi} \sum_i m_i \ln |\mathbf{x} - \mathbf{x}_i|, \quad (3)$$

$$\mathbf{A}_g(\mathbf{x}) = -\frac{s_c^g}{2\pi c^2} \sum_i m_i \mathbf{v}_i \ln |\mathbf{x} - \mathbf{x}_i|, \quad (4)$$

$$\phi(\mathbf{x}) = -\frac{s_c^{em}}{2\pi} \sum_i q_i \ln |\mathbf{x} - \mathbf{x}_i|, \quad (5)$$

$$\mathbf{A}(\mathbf{x}) = -\frac{s_c^{em}}{2\pi c^2} \sum_i q_i \mathbf{v}_i \ln |\mathbf{x} - \mathbf{x}_i|, \quad (6)$$

where $s_c^g = -4\pi G_{\text{eff}}$ and $s_c^{em} = +4\pi k_e$ are the source coefficients from §9.2, and $\mathbf{v}_i = \mathbf{p}_i/(\gamma_i m_i)$. These follow from the 2D fundamental solution of ∇^2 , $G(r) = (2\pi)^{-1} \ln r$, applied to point-source charge/mass and current distributions $\rho(\mathbf{y}) = \sum_i Q_i \delta(\mathbf{y} - \mathbf{x}_i)$. The velocity moves from the scalar to the vector exactly as in the 3D Liénard-Wiechert form; in the static-source limit the scalar parts dominate and the vector parts are an $\mathcal{O}(v/c)$ correction.

Sub-cell clip

The logarithm diverges at $\mathbf{x} = \mathbf{x}_i$. Clip the radius at $r_{\min} = \frac{1}{2} dx$ (smaller than any cell-particle separation that can actually occur on the half-cell-centered deposition lattice) when evaluating the analytic sum; the resulting value at the source cell is finite and within the floating-point dynamic range of all nearby cells. The clipped value is not the discrete 5-point Poisson solution – see §3 – but the residual mismatch is handled by the friction absorber introduced below.

Boundary-mean subtraction (supersedes the taper)

Revision 2 (2026-06-05). An earlier draft of this addendum multiplied the L-W field by a smooth taper $T(d)$ to bring it to zero through the absorbing zone. That taper is *not* gauge-clean – multiplying the field distorts its gradient (hence the force) inside the ring – and it is no longer used. It is replaced by the boundary-mean subtraction below.

Under zero-Dirichlet the bare L-W field is large and *positive* at the wall: the 2D log *grows* outward, so $\Phi(\text{wall}) \sim -(s_c/2\pi) m \ln R_{\text{wall}}$, which for the binary exceeds the near-field well depth by a factor ~ 7 –15. Forcing that to zero is the dominant startup transient.

The fix is a pure *gauge* shift: for each particle i , subtract the mean of $\ln r$ over the boundary ring from that particle's contribution before accumulating it,

$$\ln |\mathbf{x} - \mathbf{x}_i| \longrightarrow \ln |\mathbf{x} - \mathbf{x}_i| - \langle \ln |\mathbf{x}_b - \mathbf{x}_i| \rangle_{b \in \partial\Omega}, \quad (7)$$

applied uniformly to all six fields (the scalar shift maps to $\mathbf{A}_g, \mathbf{A}$ through the same $\mathbf{v}/c^2$ factor). Because it is a constant per particle, it changes no gradient and hence no force anywhere. The summed field then has zero mean on the boundary ring – consistent with the Dirichlet fixed point – removing $\sim 93\%$ of the boundary mismatch (the residual is the $\sim 7\%$ spatial variation of $\ln r$ around the ring, $\ln \sqrt{2}/\ln R_{\text{wall}}$, which a constant cannot remove). Under Neumann BC the subtraction is harmless but unnecessary.

Frozen-particle friction settle

The boundary-mean L-W is a close *initial guess* but still differs from the discrete fixed point by the near-source discrete-vs-continuum mismatch (§3, $\sim 4\%$ peak). Relax it to the fixed point by iterating the field with the particles *frozen* and the friction absorber on (`gr_sim_init_potentials_settled`): positions and velocities are pinned, so the sources are constant and the wave equation – now genuinely dissipative thanks to the friction of §4 – converges to the static solution. This is the v38 §15.9 convergence iteration, which *only works now* that friction damps the lowest mode (§4); the close L-W guess makes it converge in a few hundred steps.

Measured (256×256 `pic.binary`): the symmetric binary’s quarter-orbit total-momentum impulse falls from 2.3×10^{-3} of mv_{orb} with no settle to $\sim 3 \times 10^{-5}$ (round-off) with $\gtrsim 100$ settle steps. Wall-ramp-only friction (no interior floor) settles equally well, so the settle does *not* require an interior friction floor – see the caution in §4.

Leapfrog initialization at $t = 0$

After the L-W direct sum is written into the curr buffer of each field, also write it into the prev and next buffers:

$$\Phi_k^{-1} = \Phi_k^0 = \Phi_k^1 = \Phi_k^{\text{LW}}. \quad (8)$$

This makes the discrete time derivative encoded by the leapfrog ($\Phi^0 - \Phi^{-1}$) exactly zero, so no spurious transient is launched by the time-integration startup. Particle momenta are set directly from the scenario initial conditions; the v38 §15.9 half-step back-extrapolation is not needed because the field is already at the static solution.

Compute cost

For N_p particles on a $W \times H$ grid the L-W sum is $\mathcal{O}(N_p \cdot W \cdot H)$ log evaluations. At $W = H = 384$, $N_p = 2$ this is $\sim 3 \times 10^5$ operations – sub-millisecond. For $N_p \gtrsim 10^4$ a tree-code or FFT convolution would be needed, but the regime the sandbox addresses (a few orbiting test masses) does not require it.

3 Continuum–discrete Poisson mismatch

The L-W direct sum is the analytic Poisson solution of a sum of Dirac delta sources. The simulation’s actual source is not a delta; it is the TSC-spread (or BUMP, or CIC) deposition kernel evaluated at the particle’s sub-cell position. Inserting the analytic L-W into the discrete 5-point leapfrog and evaluating the residual $\nabla_5^2 \Phi^{\text{LW}} + s_c \rho^{\text{TSC}}$ shows that the two are not equal at the source. At a single particle of mass m centered exactly on a cell, with TSC weights (9/16, 3/32, 1/64) on the 3×3 stamp:

$$\nabla_5^2 \Phi^{\text{LW}}|_{\text{center}} = -\frac{4 \Phi_{\text{center}}^{\text{LW}}}{dx^2} \approx +5.55 \times 10^{-2} \quad (\text{at } m = 0.01, G_{\text{eff}} = 1), \quad (9)$$

while

$$s_c \rho_{\text{center}}^{\text{TSC}} = -4\pi G_{\text{eff}} \frac{m \cdot 9/16}{dx^2} \approx -7.07 \times 10^{-2}. \quad (10)$$

The residual -1.5×10^{-2} is roughly 20% of $|s_c \rho|$ at the source cell. It produces a per-step kick on Φ_g at the source of $c^2 dt^2 \times (-1.5 \times 10^{-2}) \approx -8 \times 10^{-3} \sim 5\%$ of $|\Phi_g^{\text{LW}}|_{\text{center}}$ every step. This is the source-term forcing that the friction absorber (next section) must dissipate.

Why not solve the discrete 5-point Poisson exactly? An SOR sweep on $\nabla_5^2 \Phi = -s_c \rho^{\text{TSC}}$ converges in $\mathcal{O}(\max(W, H))$ iterations to the bit-exact discrete Poisson solution and removes this residual entirely. This was implemented and tested. However, the SOR solution requires a boundary condition: zero-Dirichlet at the outer ring is the natural choice, but the resulting static field is the Dirichlet-Poisson solution of the box, not the free-space 2D log. The two differ visibly near the wall, where the Dirichlet enforcement forces $\Phi \rightarrow 0$ rather than continuing the $\ln r$ tail. For the pedagogical sandbox the free-space appearance of L-W is preferable, so we accept the small discrete-vs-continuum mismatch and rely on the friction absorber to dissipate the resulting per-step forcing.

4 Derivative-friction absorber

Replace the multiplicative damping ring of §9.6 with a derivative- friction term added to the wave equation:

$$\frac{\partial^2 \Phi}{\partial t^2} + 2\gamma(\mathbf{x}) \frac{\partial \Phi}{\partial t} = c^2 \nabla^2 \Phi + s_c \rho. \quad (11)$$

The crucial difference from the multiplicative form is that the friction term vanishes identically at $\partial_t \Phi = 0$, so the static fixed point is exactly the Poisson solution regardless of the spatial profile of γ . In particular: a uniform friction floor γ_0 everywhere in the box damps all modes – including the lowest standing mode, which has nonzero $\partial_t \Phi$ even though its spatial amplitude at the wall is small – without shifting the *static* equilibrium.

Caveat (Revision 2): “damps all modes with nonzero $\partial_t \Phi$ ” includes the legitimate, time-varying near-field of *moving* sources. A uniform interior floor therefore bleeds real orbital energy (numerical inspiral), so for dynamics the floor is set to zero and the absorber is wall-ramp-only; see §4 “Per-cell profile”.

4.1 Mode-by-mode decay rate

Consider a single eigenmode $\Phi_k(\mathbf{x}, t) = a_k(t) \phi_k(\mathbf{x})$ of the spatial operator $-c^2 \nabla^2$ with eigenvalue ω_k^2 , and let $\bar{2}\gamma_k$ denote the eigenmode-averaged friction. In the source-free regime:

$$\ddot{a}_k + \bar{2}\gamma_k \dot{a}_k + \omega_k^2 a_k = 0 \implies a_k(t) \propto e^{-\bar{\gamma}_k t} \cos(\sqrt{\omega_k^2 - \bar{\gamma}_k^2} t + \delta). \quad (12)$$

For $\bar{\gamma}_k < \omega_k$ (the underdamped regime) every mode decays at rate $\bar{\gamma}_k$. The lowest mode’s eigenmode-averaged friction is $\bar{\gamma}_{(1,1)} \sim \gamma_0$ (dominated by the uniform floor, since the wall ramp contributes only at the small-amplitude boundary). A uniform $\gamma_0 dt = 10^{-3}$ gives mode-amplitude decay to $1/e$ in ~ 1000 steps. Higher- k modes are also damped by the wall ramp; combined decay rates approach γ_{wall} for modes localized near the boundary.

4.2 Discrete leapfrog: post-step bleed (UNSTABLE)

The natural-looking discretization of (11) is to run the standard leapfrog and then, after each rotation, bleed the prev buffer toward curr:

$$\Phi_{k,\text{bled}}^{n-1} := (1 - \alpha_k) \Phi_k^{n-1} + \alpha_k \Phi_k^n, \quad \alpha_k = 2\gamma_k dt. \quad (13)$$

This appears to reduce the encoded time derivative $\Phi^n - \Phi^{n-1}$ by a factor $(1 - \alpha_k)$ each step without touching the static fixed point (where $\Phi^n = \Phi^{n-1}$ and the bleed is a no-op).

This form is unconditionally unstable at the 2D Nyquist mode for any $\alpha > 0$ when CFL = $1/\sqrt{2}$.

The recurrence implied by (13) is

$$\Phi^{n+2} = (2 - \alpha - s^2) \Phi^{n+1} - (1 - \alpha) \Phi^n + c^2 dt^2 \cdot (\text{source terms}), \quad (14)$$

where $s^2 = c^2 dt^2 |\nabla_5^2|$. At the 2D Nyquist ($s^2 = 4$ when $\text{CFL} = 1/\sqrt{2}$), the characteristic equation $\lambda^2 - (2 - \alpha - s^2)\lambda + (1 - \alpha) = 0$ has discriminant $\alpha^2 + 8\alpha > 0$ for any $\alpha > 0$. The roots are

$$\lambda_{\pm} = \frac{-2 - \alpha \pm \sqrt{\alpha^2 + 8\alpha}}{2}. \quad (15)$$

For $\alpha = 10^{-3}$, $\lambda_- \approx -1.045$ – the Nyquist mode grows 4.5% per step. Round-off seeds it; saturation in ~ 100 steps.

Empirically (the failure mode that prompted this addendum): on a 384×384 grid with L-W initialization and post-step-bleed friction at $\alpha = 0.04$ near the wall, a coherent wavefront appears at the boundary within a few hundred steps, propagates inward, and dominates the entire field within a few thousand. The “absorbing layer” is acting as a generator.

4.3 Centered-implicit discretization (STABLE)

Discretize the friction term at the leapfrog midpoint, so the friction gradient term $\partial_t \Phi$ uses $(\Phi^{n+1} - \Phi^{n-1})/(2dt)$:

$$\frac{\Phi^{n+1} - 2\Phi^n + \Phi^{n-1}}{dt^2} + 2\gamma \frac{\Phi^{n+1} - \Phi^{n-1}}{2dt} = c^2 \nabla_5^2 \Phi^n + s_c \rho^n. \quad (16)$$

Solving for Φ^{n+1} with $\beta = \gamma dt$:

$$(1 + \beta) \Phi^{n+1} = 2\Phi^n - (1 - \beta) \Phi^{n-1} + dt^2 (c^2 \nabla_5^2 \Phi^n + s_c \rho^n). \quad (17)$$

At the static fixed point $\Phi^{n+1} = \Phi^n = \Phi^{n-1}$ the LHS factor $(1 + \beta)$ and the RHS coefficient $-(1 - \beta) + 2 = 1 + \beta$ match identically – the static is preserved.

Stability at 2D Nyquist. The characteristic equation

$$(1 + \beta)\lambda^2 - (2 - s^2)\lambda + (1 - \beta) = 0 \quad (18)$$

at $s^2 = 4$ has discriminant $(2 - 4)^2 - 4(1 - \beta)(1 + \beta) = 4\beta^2$, giving real roots

$$\lambda_+ = -\frac{1 - \beta}{1 + \beta}, \quad \lambda_- = -1. \quad (19)$$

$|\lambda_+| < 1$ for $\beta > 0$; $|\lambda_-| = 1$. The Nyquist mode is marginally stable on one branch and damped on the other – unconditionally non-growing for any $\beta \geq 0$. At lower- k modes the discriminant is negative and $|\lambda|^2 = (1 - \beta)/(1 + \beta) < 1$ for $\beta > 0$: every interior mode strictly decays.

Post-leapfrog correction implementation. Equation (17) is equivalent to running the standard (un-friction) leapfrog and then applying a per-cell post-correction *before* the buffer rotation:

$$\Phi_k^{n+1} \leftarrow \frac{\Phi_{k,\text{std}}^{n+1} + \beta_k \Phi_k^{n-1}}{1 + \beta_k}. \quad (20)$$

This is the form actually used in the implementation: no leapfrog kernel needs to be modified, only a single $\mathcal{O}(N_{\text{cells}})$ sweep per field per step. At $\beta_k = 0$ the formula reduces to $\Phi_{k,\text{std}}^{n+1}$ identically, preserving backward compatibility for tests that do not use the friction array.

4.4 Per-cell profile

The implementation stores $\beta_k = \gamma_k dt$ as a single floating-point array `friction_d[k]`. The profile is a polynomial ramp from an interior floor to a larger wall value:

$$\beta_k = \begin{cases} \beta_{\text{floor}} & d_k \geq N_{\text{damp}} \\ \beta_{\text{floor}} + (\beta_{\text{wall}} - \beta_{\text{floor}}) u^2, u = (N_{\text{damp}} - d_k)/N_{\text{damp}} & 0 \leq d_k < N_{\text{damp}} \end{cases} \quad (21)$$

with d_k the cell-count distance from the nearest wall.

Revision 2 (2026-06-05): use $\beta_{\text{floor}} = 0$ (wall-ramp only) for dynamics. A nonzero interior floor was first chosen to damp the lowest standing mode everywhere. But friction damps *every* mode with nonzero $\partial_t \Phi$ – including the legitimate, time-varying orbital near-field of moving sources. A floor of 10^{-3} bleeds the binary’s orbital energy: a measurable numerical *inspiral* ($\sim 13\%$ orbit shrink over 4 orbits in `pic.binary`). The interior floor is moreover unnecessary: with the boundary-mean L-W initial guess, the frozen-particle settle reaches the fixed point using the *wall ramp alone* (the near-source residual radiates to the wall), and during dynamics there is no startup transient to trap. The recommended values for the web demo (384×384 , $N_{\text{damp}} = 48$) are therefore

$$\beta_{\text{floor}} = 0, \quad \beta_{\text{wall}} = 2 \times 10^{-2}. \quad (22)$$

A small floor ($\sim 10^{-4}$) is acceptable if a particular scenario excites a persistent interior mode, at the cost of a correspondingly small inspiral; the default is 0.

The wall ramp gives radiative wave packets several e-folds of attenuation across the ring; modes localized near the boundary decay in ~ 50 steps. At $\text{CFL} = 1/\sqrt{2}$ this is well inside the stability region (§4, any $\beta \geq 0$).

5 The DC mode under each boundary condition

The friction absorber damps every mode with nonzero $\partial_t \Phi$. The constant-everywhere mode $\Phi_{DC}(\mathbf{x}, t) = a(t)$ has $\partial_t a$ that is *not* zero in general – the wave equation drives it via the spatial mean of the source:

$$\ddot{a} = s_c \bar{\rho}, \quad \bar{\rho} = \frac{1}{|\Omega|} \int_{\Omega} \rho d^2x. \quad (23)$$

For a net-nonzero source $\bar{\rho} \neq 0$ (any scenario with nonzero total mass or charge), a undergoes constant acceleration: $a(t) = \frac{1}{2} s_c \bar{\rho} t^2$ plus initial conditions.

5.1 Dirichlet outer wall

Zero-Dirichlet at the outermost ring forces $a(t) \equiv 0$ at the wall, which under enforcement at every step pins the DC mode for the entire box. The friction term then sees no DC content, and the static fixed point is the discrete Dirichlet-Poisson solution. The *near-wall* value of Φ is forced to zero rather than continuing the L-W $\ln r$ tail; the L-W initialization must be tapered through the absorbing zone to match.

This is the configuration recommended for net-nonzero sources, and is what the v38 §9.6 absorber implicitly relied on (the multiplicative form does the pinning automatically).

5.2 Neumann outer wall

Zero-Neumann ($\partial_n \Phi = 0$ at the wall) leaves the DC mode unconstrained. With $\bar{\rho} \neq 0$ the DC mode runs away quadratically in time. For the demo’s `pic.binary` ($m_{\text{total}} = 0.02$, $WH = 147456$)

at $\bar{\rho} \sim 10^{-7}$ and $s_c \sim 10$, the DC component grows past $|\Phi^{\text{static}}| \sim 10^{-1}$ in $\sim 10^3$ steps – visually rapid.

Two gauge-fix options:

1. **Zero-mean each step.** After the leapfrog rotation, subtract the spatial mean of Φ_g (and ϕ_{em}) from prev and curr. This is gauge-equivalent (Poisson is defined up to a constant; forces depend on gradients only). Tested implementation: works for pic_static but exhibits a small bulk-momentum drift in multi-particle scenarios – the per-step constant subtraction apparently interacts with the centered-friction term in a way that bleeds momentum out of the conserved sector. Cause not yet isolated; flagged for follow-up.
2. **Pin one cell.** Force Φ at a single fixed cell to zero each step. Equivalent to a one-point Dirichlet condition; preserves Neumann elsewhere. Not implemented in v42.

The pragmatic choice in v42 is to use **Dirichlet outer BC** plus friction. The friction does the work the multiplicative damping was supposed to do (absorbing the lowest mode), and the Dirichlet condition does the work the multiplicative damping was *also* doing as a side effect (pinning the DC). Separation of concerns.

6 Summary of replacement algorithms

Replacement for v38 §15.9 (field initialization)

Step 1. Place all particles at their scenario IC positions and momenta.

Step 2. For each particle, evaluate Eqs. (3)-(6) at every grid cell, *subtracting that particle's boundary-ring mean of $\ln r$ from its contribution* (gauge shift; all six fields). Clip r at $\frac{1}{2}dx$ at the source. This leaves the summed field ≈ 0 on the boundary ring, consistent with Dirichlet. (No multiplicative taper – it distorts gradients.)

Step 3. Copy the resulting curr buffer into prev and next so the encoded time derivative is identically zero.

Step 4. Settle to the discrete fixed point: freeze the particles and iterate the field $\sim \max(W, H)$ steps with the friction absorber on (wall ramp suffices). This sheds the near-source discrete-vs-continuum residual; the momentum impulse on release falls to round-off. Reset `step_count` to 0.

Step 5. Release the particles and begin dynamics. No half-step momentum back-extrapolation is needed – the field is already static.

Replacement for v38 §9.6 (absorbing boundary layer)

Step 1. Allocate `friction_d[k]` of size $W \times H$.

Step 2. Populate with a *wall-ramp-only* profile: $\beta_{\text{floor}} = 0$ in the interior, rising as $\beta_{\text{wall}} u^2$ over the last N_{damp} cells from each wall. Do *not* use a nonzero interior floor for dynamics – it bleeds the orbital near-field energy of moving sources (numerical inspiral).

Step 3. In `gr_sim_step`, after the standard leapfrog computes `next` for every field but *before* the three-pointer rotation, apply the per-cell correction (20).

Step 4. Outer-wall boundary: zero-Dirichlet (`next[outer ring] = 0`) is recommended for net-nonzero sources. Neumann is available as a flag but requires an additional gauge fix;

see §5.

Step 5. No multiplicative damping is applied – the entire absorber function is carried by the friction.

7 Status & open items

What works (Revision 2, 2026-06-05). The combination *boundary-mean L-W + frozen-friction settle + Dirichlet + wall-only friction + BUMP $R = 8$ deposition + $v_{factor} = 1.05$* gives a stable, near-circular, non-inspiraling, COM-pinned binary:

- No lowest-mode trap, no inversion oscillation, no startup transient (boundary-mean + settle).
- Center of mass pinned: $\max |\mathbf{p}_0 + \mathbf{p}_1| \sim 4 \times 10^{-4} m v_{orb}$ over many orbits.
- Separation breathing $\sim 2\%$, secular growth ≈ 1.0 (no numerical inspiral), stable through ~ 8 orbits.
- Discrete static is reached and held.

Resolved since Revision 1.

- *COM drift on pic.binary* – the first draft attributed this to the L-W startup transient. Measurement (pre-settling the field to its exact fixed point did *not* reduce it) showed the dominant cause was instead the **TSC deposition kernel**. **BUMP $R = 8$** pins the COM ($\sim 3700\times$ smaller drift than TSC). Self-field subtraction is *not* needed for this.
- *Orbit breathing* – two compounding causes, both fixed: (a) the interior friction floor was bleeding orbital energy (now wall-ramp-only friction, §4); (b) a genuine discrete-vs-continuum velocity calibration – the discrete force during motion is $\sim 8\%$ stronger than the continuum $\sqrt{G_{eff}m}$, so the matched circular speed is $v_{factor} \approx 1.05$. Note the *rest*-measured force ratio (1.20) overshoots; the moving optimum must be found by minimizing breathing, because the velocity-dependent (gravitomagnetic / inductive) forces reduce the effective inward force during motion.

Open follow-ups.

- *Continuum-discrete Poisson residual*. See §3. Currently shed by the frozen-friction settle; could be eliminated at $t = 0$ by a TSC-convolved L-W.
- *Slow eccentricity creep* of the calibrated binary (breathing rises from $\sim 2\%$ at 4–8 orbits to $\sim 5\%$ by 12). Minor; likely accumulated discrete-force anisotropy.
- *Neumann + zero-mean momentum bleed*. See §5, item 1. Worked around by using Dirichlet.
- *$v_{factor} = 1.05$ is empirical* and specific to the mass / radius / kernel / grid; a general scenario would need to re-calibrate.